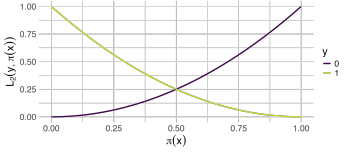
Introduction to Machine Learning

Advanced Risk Minimization
L2/L1 Loss on Probabilities



Learning goals

- Brier score / L2-Loss on probabilities
- Derivation of risk minimizer
- Optimal constant model
- L1 loss on probabilities

BRIER SCORE

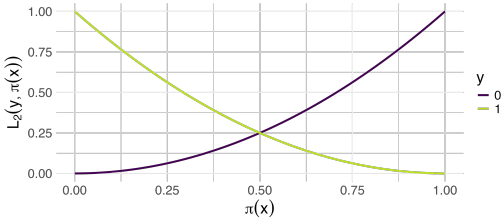
The binary Brier score is defined on probabilities $\pi(\mathbf{x}) \in [0, 1]$ and 0-1-encoded labels $y \in \{0, 1\}$ and is the $L2$ loss on probabilities.

$$L\left(y, \pi(\mathbf{x})\right) = \left(\pi(\mathbf{x}) - y\right)^2$$

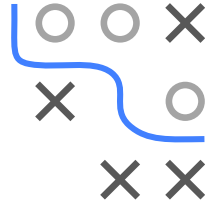
As the Brier score is a proper scoring rule (cf. section on proper scoring rules), it can be used for calibration. Despite convex in $\pi(\mathbf{x})$,

$$L\left(y, f(\mathbf{x})\right) = \left(\left(1 + \exp\left(-f(\mathbf{x})\right)\right)^{-1} - y\right)^2$$

as a composite function is not convex in $f(\mathbf{x})$ anymore (log. sigmoid for $\pi(\mathbf{x})$).



BRIER SCORE: RISK MINIMIZER



The risk minimizer for the (binary) Brier score is

$$\pi^*(\tilde{\mathbf{x}}) = \eta(\tilde{\mathbf{x}}) = \mathbb{P}(y = 1 \mid \mathbf{x} = \tilde{\mathbf{x}}),$$

which means that the Brier score attains its minimum if the prediction equals the "true" probability $\eta(\mathbf{x})$ of the outcome.

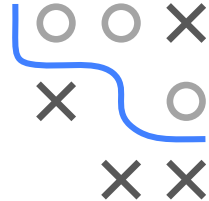
The risk minimizer for the multiclass Brier score is

$$\pi^*(\tilde{\mathbf{x}}) = \mathbb{P}(y = k \mid \mathbf{x} = \tilde{\mathbf{x}}).$$

Proof: We only show the proof for the binary case. We need to minimize

$$\mathbb{E}_x \left[L(1, \pi(\mathbf{x})) \cdot \eta(\mathbf{x}) + L(0, \pi(\mathbf{x})) \cdot (1 - \eta(\mathbf{x})) \right],$$

BRIER SCORE: OPTIMAL CONSTANT MODEL



The optimal constant probability model $\pi(\mathbf{x}) = \theta$ w.r.t. the Brier score for labels from $\mathcal{Y} = \{0, 1\}$ is:

$$\arg\min_{\theta} \mathcal{R}_{\text{emp}}(\theta) = \arg\min_{\theta} \sum_{i=1}^{n} \left(y^{(i)} - \theta\right)^2 = \frac{1}{n} \sum_{i=1}^{n} y^{(i)}$$

This is the fraction of class-1 observations in the observed data.
(This directly follows from our *L2* proof for regression).

Similarly, for the multiclass brier score the optimal constant is

$$\hat{\theta}_k = \frac{1}{n} \sum_{i=1}^{n} [y = k]$$

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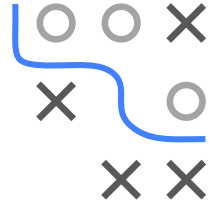
CALIBRATION AND THE BRIER SCORE



A predictor $\pi(\mathbf{x}) \in [0, 1]$ is *calibrated* if

$$\mathbb{P}\big(y = 1 \mid \pi(\mathbf{x}) = p\big) = p \quad \forall\, p \in [0, 1].$$

Intuitively, this means if we predict p, then in $100p\%$ of cases we observe $y = 1$ (neither over- or underconfident). Recall the risk minimizer for the Brier score is

$$\pi^*(\mathbf{x}) = \eta(\mathbf{x}) = \mathbb{P}(y = 1 \mid \mathbf{x}).$$

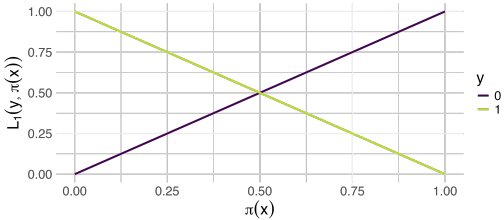
Since $\pi^*(\mathbf{x}) = \eta(\mathbf{x})$ exactly, it follows that the optimal predictor satisfies

$$\mathbb{P}\big(y = 1 \mid \pi^*(\mathbf{x}) = p\big) = p,$$

i.e., is perfectly calibrated.

L1 LOSS ON PROBABILITIES

The binary L1 loss defined on probabilities $\pi(\mathbf{x}) \in [0, 1]$ and 0-1-encoded labels $y \in \{0, 1\}$ is given by $L(y, \pi(\mathbf{x})) = |\pi(\mathbf{x}) - y|$. As the L1 loss is not a *strictly* proper scoring rule (cf. section on proper scoring rules), it should not necessarily be used for calibration. Despite convex in $\pi(\mathbf{x})$,

$$L(y, f(\mathbf{x})) = |(1 + \exp(-f(\mathbf{x})))^{-1} - y|$$

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